

torch.nn.functional.softmax#

<https://pytorch.org/docs/stable/generated/torch.nn.functional.softmax.html#t>

Description:

Apply a softmax function. Softmax is defined as:

$$\text{Softmax}(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}$$

It is applied to all slices along dim, and will re-scale them so that the elements lie in the range [0, 1] and sum to 1. See Softmax for more details.

Function Signature

```
torch.nn.functional.softmax(input, dim=None, _stacklevel=3, dtype=None)[source]#
```

Parameters

Return type

Returns:

Tensor

Notes & Warnings

NOTE

Note This function doesn't work directly with NLLLoss, which expects the Log to be computed between the Softmax and itself. Use `log_softmax` instead (it's faster and has better numerical properties).

Detailed Documentation

Documentation

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See Softmax for more details.

input (Tensor) – input

dim (int) – A dimension along which softmax will be computed.

dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

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